Progress Report: Drought Severity Levels Using Time-Series Classification

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1 Progress

Thus far, we have preprocessed the datasets used by the program. Rows with missing data were removed, and the relevant feature columns were selected. Furthermore, algorithms were designed, and a system pipeline has been established. The macro F1 score and Mean Absolute Error (MAE) were chosen as the evaluation metrics, as we can calculate the performance of the models across all classes and how severe the prediction errors are.

2 Problem Statement

Drought in the US is a highly destructive and worsening phenomenon, on average causing $9.6 billion in damages per event [1]. To mitigate and prepare for the effects of drought, predicting them early is essential. Unfortunately, drought is a complex and little-understood phenomenon, reducing the effectiveness of traditional prediction methods. Other factors than purely precipitation, such as temperature and human activities, can shape the effects of drought, necessitating a more complex model.

While machine learning models have been made for predicting drought, it is uncommon for them to utilize data about land surface characteristics, which can have large impacts on drought likelihood. Our project aims to adjust for this by using static soil data alongside various temporal meteorological data to account for most possible factors and accurately predict drought severity. The project will make use of an Autoregressive Integrated Moving Average with Exogenous variables (ARIMAX), alongside LSTM and 1D-CNN deep learning models to explore multiple effective models for our data. The goal is to create an accurate predictor of drought after comparing model performance and iteration.

3 Literature Review

Due to its importance, the concept of using machine learning for drought prediction has been studied before. Artificial Neural Networks (ANN's) are a common model, with multiple models being shown to outperform the leading drought prediction model, Autoregressive Integrated Moving Average (ARIMA). Their success can be owed to their ability to model nonlinear relations between the many relevant attributes to make accurate predictions [2]. Random Forest (RF) models, relying on multiple decision trees constructed from random subsets of the data, have also been tested [2]. As they do not rely on any preexisting assumptions about the data nor scale adjustments amongst parameters, they were found to be effective with the large datasets needed to predict drought. Many other models to predict drought have also been researched recently, including but not limited to Genetic Algorithms, Fuzzy Logic, and Support Vector Machines [3].

4 Data Collection

In this project, we will utilize a multi-source dataset, provided in the IEEE-CIS format on Kaggle, that integrates temporal meteorological observations with static geospatial as well as soil characteristics to present a comprehensive view of the factors causing a drought. Primary data is

sourced from the NASA POWER Project, which labels 18 daily meteorological indicators including the most important ones required for this project, such as precipitation (*PRETCOT*), temperature (*T2M*), and humidity (*QV2M*). USDM (U.S. Drought Monitor) is used to assess the drought severity levels for the U.S. using weekly data from ground to air. All areas of the U.S. are classified into one of 6 levels (None to D4) based on the conditions present at that time. In addition to USDM providing information regarding the drought, HWSD (Harmonized World Soil Database) provides county specific environmental context through 32 static features including elevation, slope (8 different categories), aspect direction (i.e. north, south, east, west), and land cover types (e.g. Forest - *FOR_LAND*; Water - *WAT_LAND*; and cropland - *CULT_LAND*).

The static soil features are merged with the time-series meteorological file using Federal Information Processing Series (*FIPS*) codes across 3,109 US counties. Daily weather data is then clustered into 90-day sequences to capture the collective environmental impacts required to assess droughts. Lastly, to mitigate the potential of data leakage and provide for strong evaluation efforts, the dataset has been chronologically partitioned-- with a **Training** Set of 2000-2009 (47%); a **Validation** Set of 2010-2011 (10%); and a **Testing** Set of 2012-2020 (43%). To account for potential shifts in climate patterns over time, our model selection will use an expanding window validation approach – training on 2000–2007 and validating on 2008, then training on 2000–2008 and validating on 2009, and so on – while keeping the 2012–2020 test set untouched.

5 Algorithms and Models

In this project, we will explore several machine learning as well as deep learning models to accurately capture the complex relationship between daily meteorological shifts and stationary soil properties. We will assess an initial ARIMAX model in order to set a standard for measuring how well we can assign linear feature importance. Additionally, we will explore several deep learning models, particularly Long Short-Term Memory (LSTM) networks and 1D-Convolutional neural networks (1D-CNN). These particular models are useful for processing the 90-day sequences of meteorological labels, such as precipitation (*PRECTOT*) and temperature (*T2M*), as well as static soil features such as elevation and land cover types.

5.1 ARIMAX (Baseline)

We will utilize an ARIMAX model as a baseline to establish a performance floor to avoid overfitting related to correlated meteorological predictors.

$$\Delta^d y_t = C + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \sum_{k=1}^{r} \beta_k x_{t,k} + \epsilon_t$$

ARIMAX combines three different components to make predictions. The autoregressive (AR) component represented by $\sum_{i=1}^{p} \phi_i y_{t-i}$, captures the relationship of the output and its past values, essentially modeling how past observations influence future ones. This allows the model to remember previous states of the system, such as persistent weather patterns. The integrated (I) component, $\Delta^d y_t$, applies differencing to stabilize the mean and variance, ensuring the data is suitable for statistical modeling. This transformation removes trends or seasonality that might skew the results. The moving average (MA) component addresses the influence of past forecast errors, effectively smoothing out the noise within the sequence. In the equation, this is the summation $\sum_{j=1}^{q} \theta_j \epsilon_{t-j}$, where ϵ represents the white noise error. By incorporating the past errors, the model adjusts for random shocks in the system and is able to learn from its past mistakes. The exogenous (X) term $\sum_{k=1}^{r} \beta_k x_{t,k}$, allows for the 18

meteorological features in our dataset to impact the output of the model.

ARIMAX serves as an effective baseline for drought prediction for several reasons. First, ARIMAX models are very commonly used in time series forecasting and are widely regarded as a standard benchmark. Second, ARIMAX provides an interpretable, well-understood benchmark that we can use to compare against our LSTM and 1D-CNN models. If these models can't meaningfully outperform ARIMAX, it indicates that the models are most likely overfitting to data rather than a genuine predictive improvement.

To apply the ARIMAX model on our dataset, we will treat the drought severity level as our endogenous target variable (y_t) and the NASA POWER meteorological indicators and HSWD soil characteristics as exogenous regressors (x_t).

Since the USDM data is released weekly, the daily weather sequences will be aggregated to match this frequency. To identify the optimal lag structure (d, p, q, r), we will use the Box-Jenkins methodology to ensure the model accurately captures the temporal dependencies of drought. This involves utilizing the Augmented Dickey-Fuller (ADF) test to determine the order of integration d, Analyzing Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots to estimate initial values for p and q, and Cross-Correlation Functions (CCF) to determine r. Once the structure is defined, we will use Maximum Likelihood Estimation to calculate the coefficients.

To evaluate the ARIMAX model we will use the Mean Absolute Error and the Macro F1 Mean to quantify the accuracy of the drought severity predictions compared to the USDM levels.

5.2 Long Short-Term Memory (LSTM)

Because of the rapid loss of the error signal through backpropagation over time in a standard RNN, long-term dependencies in time-series data can't be learned. To remedy this issue, the LSTM (Long Short-Term Memory) neural network architecture was created to store data over long timeframes, enabling the model to learn from historical data and make informed decisions based on that knowledge.

The core functionality of LSTM is its memory cell, which has a linear unit with a constant connection to itself known as the Constant Error Carousel, or "CEC". To avoid having this central memory cell influenced by irrelevant daily changes in the weather, the CEC allows information to flow while preventing crucial error signals from vanishing or exploding through the use of multiplicative gate units: the input gate and the output gate.

Furthermore, when an LSTM receives an input, the shape of the input will be a matrix of 90 timesteps times 18 features. For each of these time steps, the network updates the input gate activation y^{in_j}, the output gate activation y^{out_j}, the internal cell state s_{cj}, and the final cell output y^{cj}, which are defined in the following:

$$y^{in_j}(t) = f_{in_j}(net_{in_j}(t))$$

$$y^{out_j}(t) = f_{out_j}(net_{out_j}(t))$$

$$s_{c_j}(t) = s_{c_j}(t-1) + y^{in_j}(t)g(net_{c_j}(t))$$

$$y^{c_j}(t) = y^{out_j}(t)h(s_{c_j}(t))$$

Central to this architecture is the $net(t)$ function, which finds out what is in the raw input by determining weighted linear combinations of the day's sensor readings plus the prior day's hidden state. After determining the raw net score, it will go through the whole network via the activation functions. The input f_{in_j} and output gates f_{out_j} in this model are logistic sigmoid functions that help scale the new weather sensor measurements to be

between 0 and 1. This helps determine how much new weather information will be added to an internal memory and how much memory related to drought will be revealed through the softmax function. The function g and h in this architecture are both differentiable functions that will enable input values and the memory to remain numerically stable. The most important part of this architecture is the CEC, which carries forward the most recent memory $s_{c_j}(t-1)$ with a constant weight of 1.0. This allows the essential long-term information to remain preserved through the entire 90-day period without exponential decay. After the 90-day sequence is fully processed, the final hidden representation is forwarded to a fully connected layer and a softmax to predict drought severity class in each respective category.

5.3 1D-Convolutional Neural Networks (1D-CNN)

A challenge with detecting patterns across timeframes for traditional RNNs is when "signals" can be present and transformed across the timeseries. These can be a sudden increase in maximum temperature (*T2M_MAX*) or long-term decreases in precipitation (*PRECTOT*), which can occur at any time. To solve this, we will use a 1D-Convolutional Neural Network (1D-CNN), a model that can detect signals regardless of position through a sliding filter.

The main component of 1D-CNN's is the convolutional layer. For each timestep, the input will be the 18 meteorological features, and thus the input will be of size $t \times 18$. Each neuron in a convolutional layer contains a kernel of learnable weights of size $k \times 18$, where k is a hyperparameter for the 18 inputs. Given a stride of s, forward propagation creates output of size $n - k \times 18$ by the function

$$y_i = b_i + \sum_{j=0}^{k-1} \sum_{l=0}^{18-1} w_{j,k} \cdot x_{i \cdot s,+i,l} \ .$$

This output will then go through the ReLU activation function $y_i = max(0, x_i)$ for non-linear learning. Output is then pooled by the MaxPooling function to aggregate output and reduce dimensions. These three layers are then repeated such that the deep learning model can learn more complex patterns. The last layers are a global pooling layer to flatten output into a 1-D vector of size 18, a fully connected (dense) layer, with the function $y = Wx + b$, where W is of size 18×2 and b is of size 2. These outputs are then used in a softmax classifier to determine the probabilities of a drought versus not with the function

$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=0}^{1} e^{z_j}}$. Results are then passed to a softmax classifier, which produces probabilities for both drought and non-drought outputs. Afterwards, 1D-CNN can use local temporal patterns in weather data as input to classify events into their own respective classes.

6 Evaluation

Given the imbalance in our meteorological dataset, using standard accuracy measurements will not work. Therefore, we will use two different types of frameworks to evaluate our models: Macro F1-Score and Mean Absolute Error. The goal is to utilize the Macro-F1 Score to ensure the model maintains high precision and recall across all classification categories. We will also use the Mean Absolute Error (MAE) to penalize "near-misses" less harshly than catastrophic misclassification. To account for potential shifts in climate patterns across the test period, we will also evaluate results across three separate 3-year time windows within the 2012–2020 test set. This segmented evaluation will help reveal whether model performance degrades or improves over time, and will show how well each model generalizes under evolving climate conditions.

7 Timeline

The project will be completed by April 21. Due to slight difficulties with organization over break and studying, our timeline has been delayed and now begins at the fourth week. The fourth week was dedicated to preprocessing, where we synchronized all of the daily weather variables and did z-score normalization on all of the different weather features for the training set. Throughout weeks five to six, we will establish baselines by implementing the ARIMAX algorithm. The core of the project will be developed between weeks seven and eight, where we will build and train the LSTM and 1D-CNN architectures using 90-day and 180-day temporal windows. Lastly, weeks nine and ten are reserved for testing and hyperparameter tuning, and synthesizing our findings into an ACM-formatted technical report and presentation.

8 Expected Results

By the end of the project, we expect to develop a working drought prediction framework that accurately identifies and categorizes drought severity levels. We expect that the 32 static soil features alongside the 90-day temporal meteorological sequences will allow the models to capture the relationship between land surface characteristics and atmospheric shifts. We anticipate that the deep learning models will be able to meet or exceed the ARIMAX model baseline MAE and Macro F1-Score.

References

[1] *The High Cost of Drought | January 23, 2020 | Drought.gov*. (2020, January 23). https://www.drought.gov/news/high-cost-drought

[2] Nandgude, N., Singh, T. P., Nandgude, S., & Tiwari, M. (2023). Drought Prediction: A Comprehensive Review of Different Drought Prediction Models and Adopted Technologies. *Sustainability*, *15*(15). https://doi.org/10.3390/su151511684

[3] Hao, Z., Singh, V. P., & Xia, Y. (2018). Seasonal Drought Prediction: Advances, Challenges, and Future Prospects. *Reviews of Geophysics*, *56*(1), 108–141. https://doi.org/10.1002/2016RG000549